BVS
Test for Degrees of Protection
IP65

Test Report **BVSPS24348**
dated 02.12.2010

DEKRA EXAM GmbH
Fachstelle für
Sicherheit elektrischer
Betriebsmittel - BVS

Carl-Beyling-Haus
Dinnendahlstraße 9
44809 Bochum

Job identification number: **20101180**	**SD number:** **32000382**	**Expert:** **Ha**

Test specimen	**: Enclosure**
Applicant	**: Elektro Mechanik GmbH**
Manufacturer	**: Elektro Mechanik GmbH**
Type designation	**: Aufbaugehäuse Typ UNI-O/K**
Type of protection	**: IP65**
Date of order	**: 30.11.2010**
BVS reg. number	**: 404/10**
Device number	**: -**
Drawing number	**: -**

Test for degrees of protection IP65 according to IEC 60529 V2.1: 2007

The enclosure with the BVS-PMNo.: 404/10 (see Fig. 1-2), was submitted to a test for degrees of protection IP65.

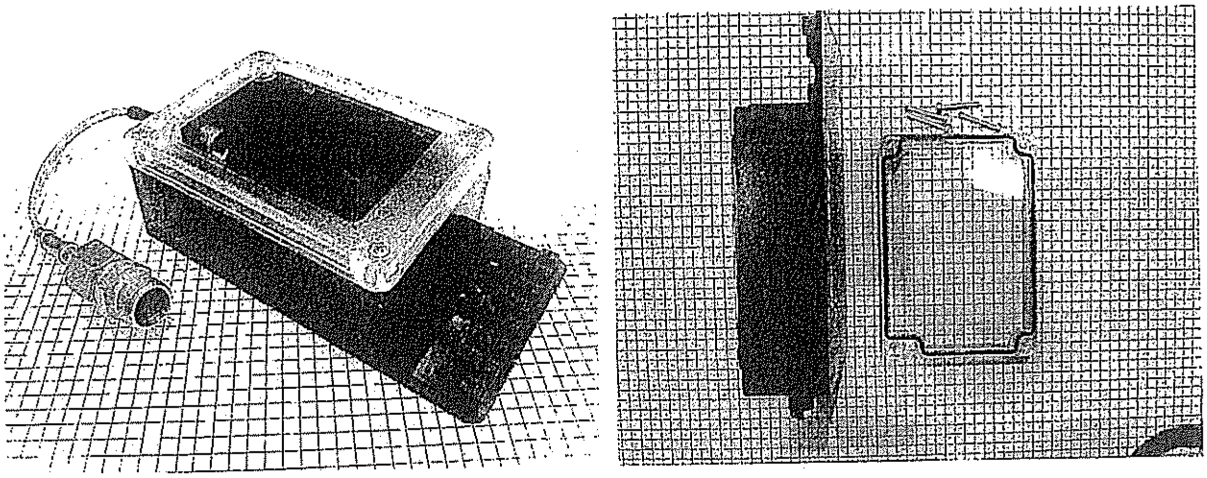

Fig. 1-2: test specimen

Testing officer: W. Kirmes

Test date: 30.11.-01.12.2010

Test description:

Test for degree of protection IP6X

Test date: 30.11.-01.12.2010
Enclosure volume: ca. 0.3 l
Low pressure: 20 mbar
Test duration: 8 hrs

Test for degree of protection IPX5

Test date: 01.12.2010
Test temperature: 18°C
Water temperature: 17°C
Test duration: 3 min
Water flow rate: 12,5 l / min

Test result:

No intrusion of dust or water was detected inside the test specimen.

The test result solely refers to the above-mentioned specimen.

Equipment used:

Dust testing chamber	IPY-07	calibrated up to 10/11
Spray sprinkler	IPX-05	calibrated up to 04/11
Timer	TM-002	calibrated up to 03/11

DEKRA EXAM GmbH
Fachstelle für Sicherheit elektrischer Betriebsmittel
Bergbau-Versuchsstrecke

02.12.2010
(Date / testing officer)

02.12.2010
(Date / for the correctness)

The translation of this test report can be relayed only complete and unchanged

Elektro-Mechanik GmbH

Product Test Certificate

Earth Fault Indicator
Type EASI

Operation test over the whole temperature range

Test description:
The operability of three examination objects is tested in the temperature range between -20°C (-4°F) and 70°C (158°F). The test has been carried out in accordance with manufaturers instructions based on: IEEE 495-2007

Serial number		Current consumption	LED indication	Flag indication	Remote indication contacts	Remote reset	External blinking lamp
-20°C -4°F	0544969	2.2 µA	passed	passed	passed	passed	passed
	0544966	2.1 µA	passed	passed	passed	passed	passed
	0544967	2.2 µA	passed	passed	passed	passed	passed
20°C 68°F	0544969	3.3 µA	passed	passed	passed	passed	passed
	0544966	2.9 µA	passed	passed	passed	passed	passed
	0544967	2.8 µA	passed	passed	passed	passed	passed
70°C 158°F	0544969	7.2 µA	passed	passed	passed	passed	passed
	0544966	6.0 µA	passed	passed	passed	passed	passed
	0544967	7.4 µA	passed	passed	passed	passed	passed

The examination object has passed the test.

Date: 24.10.2008

signature, stamp
Elektro-Mechanik GmbH, Velbert

signature, stamp
ABB AG, Ratingen, Germany

Elektro-Mechanik GmbH

Product Test Certificate

Earth Fault Indicator
Type EASI

Temperature Test

Test description:
Three examination objects are tested in the temperature range between -20°C (-4°F) and 70°C (158°F). The objective is to show that the temperature-dependent variance of the trip current is less than +/- 10% of the nominal trip current. The test has been carried out in accordance with manufaturers instructions based on: IEEE 495-2007

Trip current			**Limit values**		**Serial number**		
			- 10%	**+ 10%**	**0544969**	**0544966**	**0544967**
First trip current nominal value: 25 A	-20°C	-4°F	**22,5 A**	**27,5**	passed	passed	passed
	20°C	68°F			passed	passed	passed
	70°C	158°F			passed	passed	passed
Second trip current nominal value: 50 A	-20°C	-4°F	**45 A**	**55 A**	passed	passed	passed
	20°C	68°F			passed	passed	passed
	70°C	158°F			passed	passed	passed
Third trip current nominal value: 125 A	-20°C	-4°F	**112,5 A**	**137,5 A**	passed	passed	passed
	20°C	68°F			passed	passed	passed
	70°C	158°F			passed	passed	passed

The examination object has passed the test.

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Product Test Certificate

Earth Fault Indicator
Type EASI

High Voltage Resistance

Test description:
The objective is to test the earth fault indicator's resistance against high voltage fields. For this, the examination object is placed 0,1 m next to a high voltage conductor. In a distance of 0,2 m beside the conductor, there is a grounded conducting plate. The high voltage conductor is connected to an AC voltage source with an amplitude of Û=20 kV. The test has been carried out in accordance with manufaturers instructions based on: IEEE 495-2007

Examination object in standby mode:
The examination object is switched to standby mode. When the Û=20 kV voltage source is switched on, the examination object shall not switch to indication mode.

Examination object in indication mode:
The examination object is switched to indication mode. When the Û=20 kV voltage source is switched on, the examination object shall not switch to standby mode.

Serial number	**High voltage resistance**	
	Sample in indication mode	**Sample in standby mode**
0544966	passed	passed
0544969	passed	passed
0544967	passed	passed

The examination object has passed the test.

Date: 24.10.2008

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Product Test Certificate

Earth Fault Indicator
Type EASI

Reset function test

Test description:
Every possibility to perform a reset has to be tested separately. The test has been carried out in accordance with manufaturers instructions based on: IEEE 495-2007

Remote reset:
The examination object is provided with a contact to perform a remote reset. In order to test this function, the examination object has to be switched from standby mode into indication mode. The examination object has to reset into standby mode, when the remote reset contacts are short circuited with a 15 Ω resistor.
Reset over operational current:
The examination object has to switch into standby mode, as soon as there is a current flow of less than 10 % of the adjusted trip current in the monitored conductor.

Serial number	Remote reset	Reset over operational current
0544966	passed	passed
0544969	passed	passed
0544967	passed	passed

The examination object has passed the test.

Date: 24.10.2008

Elektro-Mechanik GmbH
D-42553 Velbert

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Product Test Certificate

Earth Fault Indicator
Type EASI

Vibration and Impact test

Test description:
The objective is to test the mechanical stability of the indicator. The test has been carried out in accordance with manufaturers instructions based on: IEEE 495-2007

Vibration test:
Three examination objects are fixed on a vibration machine. The vibration machine has a stroke of 2 mm and a frequency of 25 Hz. After 2 minutes the examination objects are checked for mechanical damages.

Impact test:
The examination objects are dropped from a height of 10 cm on a flat surface. After this, the examination objects are checked for mechanical damages.

Serial number	**Mechanical inspection**	
	Vibration test	**Impact test**
0544966	passed	passed
0544967	passed	passed
0544969	passed	passed

The examination object has passed the test.

Date: 24.10.2008

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Product Test Certificate

Earth Fault Indicator
Type EASI

Tensile force test of the cable and connectors

Test description:
The objective is to test the mechanical stability of the cable and the connector.
A tensile force of 13,6 kg (30 lb) is pulling at the cable which is connected to the screwing terminal of the indicator. After one hour the examination objects are checked for mechanical damages. The test has been carried out in accordance with manufaturers instructions based on: IEEE 495-2007

Seriennummer	Mechanical inspection
0544971	passed
0544941	passed
0544967	passed

The examination object has passed the test.

Date: 24.10.2008

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Product Test Certificate

Earth Fault Indicator
Type EASI

Continuous Current and High Current Pulse Test

Test description:
The examination objects have to resist a continuous operating current and high current pulses. The test has been carried out in accordance with manufaturers instructions based on: IEEE 495-2007

Continuous current test:
The examination objects are connected to a current of 240 A for a time of 60 seconds. After this, all (relevant) components are checked for a temperature increase.

High current pulse test:
The examination object has to be adjusted to an average trip current. Before the test begins, this trip current has to be measured. Now, five 160 ms current pulses are performed with an amplitude of 5000 A. After the high current pulses, the trip current has to be measured again. There should be no difference (smaller than 4%) to the trip current measured before.

Serial number	Continuous test current	High current pulse test	
		Trip current (before)	Trip current (after)
0544967	passed	51 A	51 A
0544966	passed	50 A	50 A
0544969	passed	51 A	50 A

The examination object has passed the test.

Date: 24.10.2008

Wolf Huss

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